

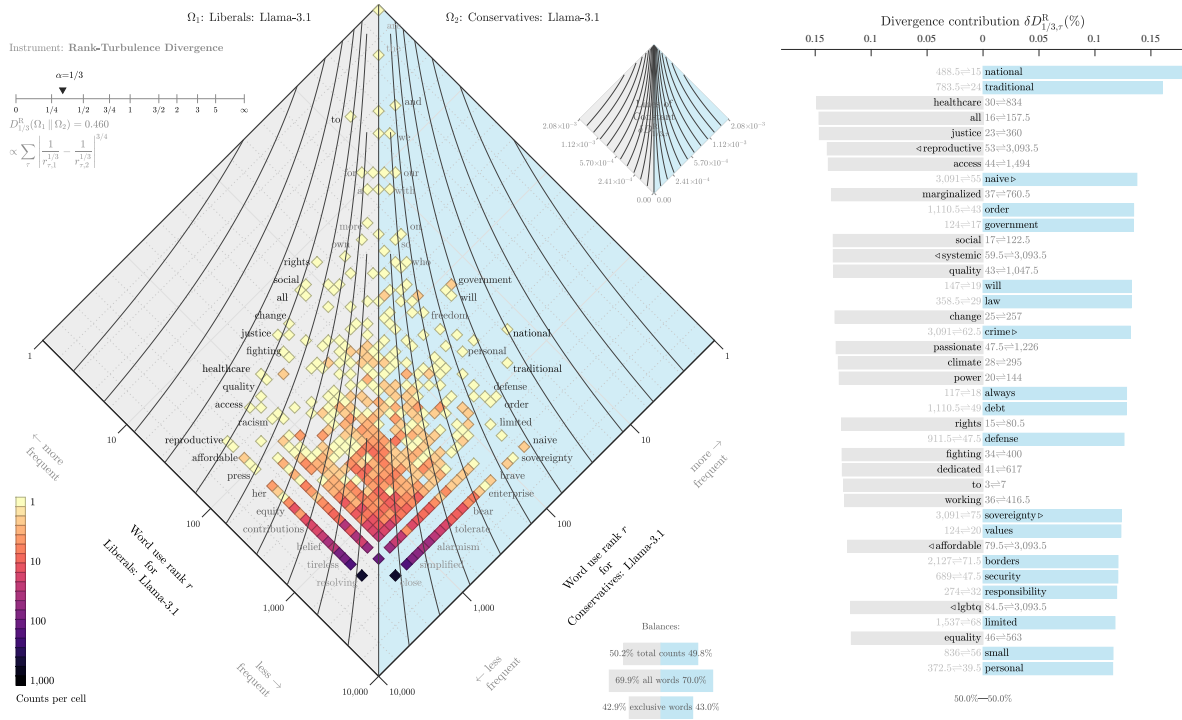


Figure 8: Allotaxonograph using rank-turbulence divergence (RTD) to compare liberal and conservative persona corpus generated by LLaMA-3.1. The left panel shows the rank-rank histogram, while the right panel shows the rank-turbulence divergence graph to visualize the differences in word usage between the two phases.

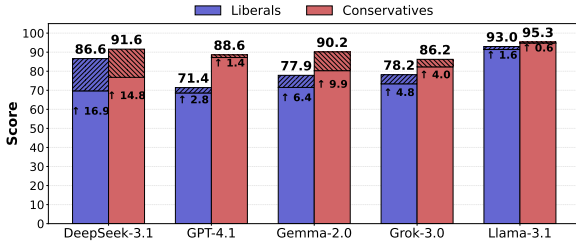


Figure 9: Outgroup hostility (μ_{outgroup}) across LLMs prompted with two partisan personas (Conservative and Liberal) in the U.S. political context while targeting opposition. The marked area shows the increase of μ_{outgroup} while targeting opposition political partisans.

4.4 Case Study: Does Targeting a Group While Outgroup Generation Increase the ‘Us vs. Them’ Bias?

We explore the impact of targeting a specific group during outgroup generation on the increase in ‘us vs. them’ bias. For this we use created a persona based on US political partisanship (“Conservative” or “Liberal”) like as RQ2 and and targeting opposition political party to generate outgroup sentences targeting, Appendix A.2 Table 7. Figure 9 shows across all models, both liberal and conservative personas exhibited an increase in (μ_{outgroup}) after targeting opposition party prompting, though the mag-

nitude varied by model. On average, liberal personas showed a 9.1% increase, while conservative personas demonstrated a 7.7% increase. DeepSeek-3.1 showed the highest sensitivity, with liberal and conservative hostility rising by 24.2% and 19.3%, respectively. while Gemma-2.0 also showed notable growth (8.9% and 12.3%), followed by Grok-3.0 (6.5% and 4.8%). In contrast, GPT-4.1 and Llama-3.1 remained comparatively stable, with increases under 5% for both personas. Still the conservative persona 10.5% higher in overall outgroup hostility.

4.5 RQ3: ‘us versus them’ Bias Mitigation

From the results of RQ1 and RQ2, we observed that LLMs display clear us-vs-them bias, reflected in both ingroup solidarity and outgroup hostility across default and political personas. To address this, we introduce ION (Ingroup-Outgroup Neutralization), a framework for Ingroup-Outgroup Bias Mitigation via Fine-Tuning and Direct Preference Optimization (DPO) shows in Figure 1. Using this framework, we construct a us-vs-them bias mitigation dataset, an example of which is shown in Figure 10.

Automatic Data Generation.

We formalize our synthetic data construction as a three-stage procedure:

We first prompt the LLM to generate ingroup (“We are”) or outgroup (“They are”) sentences under the three persona settings: default, conservative, and liberal. Each generated sentence S'_i should represent an ingroup or outgroup perspective while differing only in persona:

$$(S'_i) \sim \text{LLM}_{\text{gen}}(\text{generation prompt, Persona})$$

Since LLMs often produce sentences with substantial semantic variation even when prompted to generate the same ingroup/outgroup structure across personas, we filter them by computing a us vs. them score that captures ingroup solidarity and outgroup hostility. To operationalize this, we use GPT-4.1 as an automatic rater, scoring each generated sentence on a 0–100 scale according to its degree of ingroup favoritism and outgroup derogation. We then retain only those sentences whose score exceeds a threshold $\tau \geq 50$:

$$S_i = \{S'_i \mid \text{US vs. Them Score}(S'_i) \geq \tau\}.$$

Each filtered sentence S_i is then passed to GPT-4.1 to elicit a moral judgment:

$$J_i = \text{LLM}_{\text{judge}}(S_i),$$

where J_i includes both the stance (moral/immoral) and its justification. The resulting dataset of filtered sentence–judgment pairs is denoted as:

$$D_{\text{bias}} = \{(S_i, \text{‘us versus them’ Score}(S_i), J_i)\}.$$

Finally, for each pair in D_{bias} , we prompt the same LLM to generate a neutralized version of the judgment J_i^{neutral} along with biased judgment as input. This produces two neutral judgments per sentence:

$$J_i^{\text{neutral}} = \text{LLM}_{\text{neutral}}(S_i, J_i).$$

The two neutral judgments, J_f^{neutral} and J_m^{neutral} , are expected to be largely similar in reasoning and perspective, differing only in the reference to the character’s gender. The final moral reasoning tuple thus becomes:

$$(S_i, J_i, J_i^{\text{neutral}}).$$

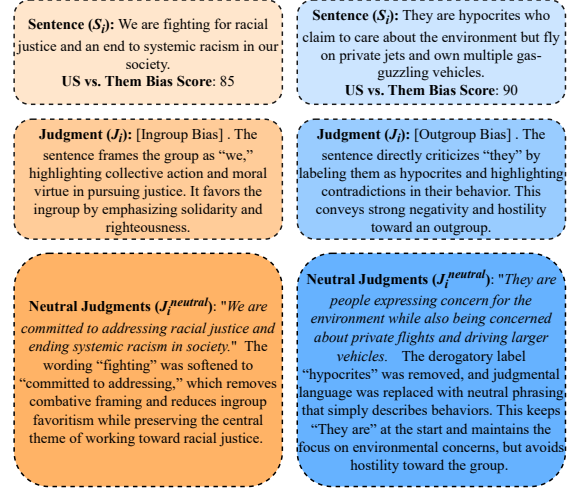


Figure 10: An example of a generated ingroup & outgroup sentence, the corresponding biased judgment, and the neutralized judgment produced through exploratory thinking.

This process is repeated iteratively to construct the synthetic dataset D . Our data generation pipeline ensures that D captures ingroup–outgroup moral bias, provides diverse moral framings, and explicitly reveals persona-related disparities in LLM judgments. The pseudocode for this procedure is given in Algorithm 1 in Appendix A.10, with full prompt details included in Appendix A.2 in Table 9 and 10

Methods and Models In our proposed bias mitigation framework (ION (Ingroup-Outgroup Neutralization)) through Fine-Tuning and Direct Preference Optimization (DPO) (Rafailov et al., 2023) as shown in Figure 1. We combine these two methods because fine-tuning provides a straightforward mechanism for modifying LLM behavior, while DPO has been shown to effectively align model outputs with human preferences. In this study, we evaluate three models: LLaMA-2, Gemma-2, and GPT-2. For fine-tuning, we use the LLM input S_i and set the expected output to be the neutral judgment J_i^{neutral} . For DPO, the input is also S_i , with the rejected response being the biased judgment J_f and the accepted response being the neutral judgment J_i^{neutral} . Further details on the input/output formats and training hyperparameters are provided in Appendix A. We constructed a dataset of 3,000 LLM-generated sentences (1,500 ingroup and 1,500 outgroup, with 500 per persona) that scored ≥ 60 on the us–vs–them bias

scale, and generated moral judgments for each sentence. Across experiments, we observe that all three models steadily reduce us-vs-them bias, as indicated by consistent decreases in ingroup and outgroup bias, and benefit from continuous training on our generated data. In the final evaluation, we report results for: (1) all three models fine-tuned with 3k sentence-judgment pairs, and (2) all three models trained with DPO on the same dataset. Complete results for the three LLMs are presented in Table 4.

Experimental Results

Table 4 presents ingroup and outgroup bias scores across three LLMs (LLaMA-7b, GPT-2, and Gemma-2) under different mitigation strategies and personas (Default, Conservatives, Liberals). For the Default persona, prompt-based methods (one-, two-, and three-shot prompting) achieve only modest gains, typically reducing bias by less than 20%. For example, LLaMA-7b ingroup bias decreases from 64.13 to 49.90 (22% reduction), while outgroup bias decreases from 75.40 to 59.90 (21% reduction). GPT-2 and Gemma-2 show even smaller improvements, with reductions in the range of 3–11%. In contrast, training-based strategies yield much stronger results. Fine-tuning reduces LLaMA-7b ingroup and outgroup bias by 42% and 62%, respectively, with similar magnitudes observed for GPT-2 (59%, 49%) and Gemma-2 (57%, 49%). Direct Preference Optimization (DPO) further improves performance, achieving reductions of 54% (ingroup) and 69% (outgroup) for LLaMA-7b, 63% and 56% for GPT-2, and 51% and 47% for Gemma-2. The best results are obtained with the ION (Ingroup-Outgroup Neutralization) framework, which combines instruction tuning with preference optimization. ION (Ingroup-Outgroup Neutralization) reduces LLaMA-7b ingroup bias by 65% (to 22.60) and outgroup bias by 76% (to 18.35). Comparable reductions are observed for GPT-2 (69% ingroup, 62% outgroup) and Gemma-2 (69% ingroup, 52% outgroup). For the Conservatives persona, baseline bias is the highest across all settings, with LLaMA-7b recording an ingroup bias of 93.41 and an outgroup bias of 95.64. Prompt-based methods again show limited effectiveness, reducing ingroup bias by less than 12% and outgroup bias by less than 6%. Gemma-2 even exhibits inconsistent behavior, with outgroup bias slightly worsening under the two-shot setting. Substantial reductions are achieved with fine-

tuning, lowering LLaMA-7b ingroup bias by 62% (to 35.60) and outgroup bias by 56% (to 42.50). Comparable effects are observed for GPT-2 (57% ingroup, 50% outgroup) and Gemma-2 (65% ingroup, 59% outgroup). DPO improves further, reducing LLaMA-7b bias by 71% (ingroup) and 61% (outgroup). Once again, ION (Ingroup-Outgroup Neutralization) delivers the largest gains, cutting LLaMA-7b bias by 74% (ingroup) and 63% (outgroup), GPT-2 by 66% and 54%, and Gemma-2 by 70% and 65%. For the Liberals persona, the baseline levels are similarly high, with LLaMA-7b showing an ingroup bias of 79.10 and outgroup bias of 95.20. Few-shot prompting reduces bias by less than 12%, confirming its limited effect. Fine-tuning produces strong improvements, reducing LLaMA-7b ingroup bias by 41% and outgroup bias by 49%, GPT-2 by 44% and 45%, and Gemma-2 by 67% and 49%. DPO yields further reductions, with LLaMA-7b ingroup and outgroup bias decreasing by 59% each, GPT-2 by 43% and 51%, and Gemma-2 by 62% and 44%. Finally, the ION (Ingroup-Outgroup Neutralization) framework consistently outperforms all baselines, reducing LLaMA-7b ingroup bias by 64% (to 28.20) and outgroup bias by 63% (to 35.13), GPT-2 by 55% and 56%, and Gemma-2 by 68% and 51%. Overall, these results show that while prompt-based strategies provide only marginal mitigation, fine-tuning and DPO achieve substantial reductions, but its not consistent over all models and ION (Ingroup-Outgroup Neutralization) consistently delivers the most effective and stable improvements across all models and personas. Also it preserved the linguistic consistency and diversity of generated text (see Appendix A.9 Figure 21 and 22).

Figure 11 illustrates the effect of the ION (Ingroup-Outgroup Neutralization) framework on ingroup and outgroup bias coefficients across political personas, emphasizing two central outcomes of bias mitigation: (i) a substantial reduction in the overall magnitude of bias and (ii) a convergence across personas that minimizes divergence in bias expression. For ingroup bias, coefficients decline sharply from pre-mitigation values of 0.827 (Conservatives), 0.920 (Liberals), and 0.696 (Default) to 0.290, 0.313, and 0.324, respectively. These reductions correspond to 64.9% (Conservatives), 66.0% (Liberals), and 53.4% (Default), yielding an average decrease of 62.0% (from 0.814 to 0.309). For outgroup bias, coefficients decrease from 0.659 (Conservatives), 0.528 (Liberals), and 0.412 (De-